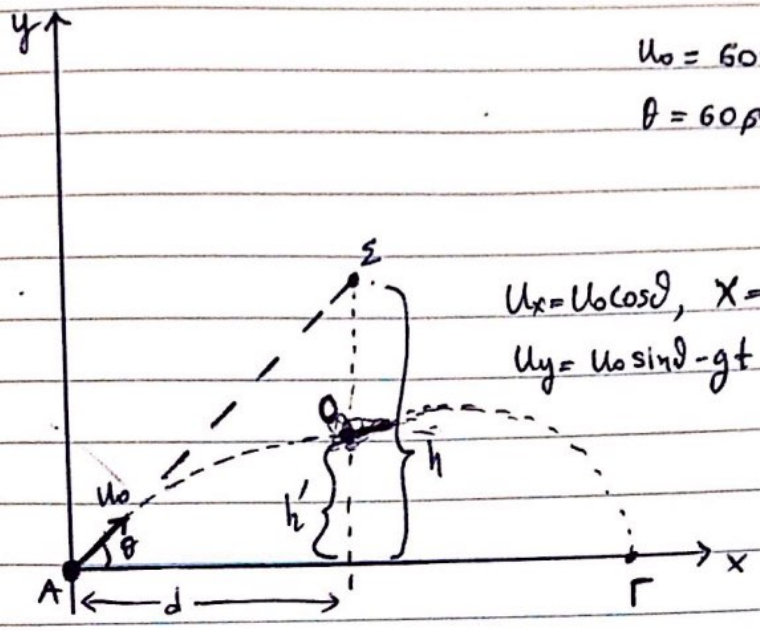
Λύσεις Ενδιάμεσης εξέτασης

Άσκηση 1



$u_0 = 60\,m/s$, $m = 2\,kg$

$\theta = 60^\circ$, $d = 60\sqrt{3}\,m$

$u_x = u_0\cos\theta$, $x = u_0\cos\theta\, t$

$u_y = u_0\sin\theta - gt$, $y = u_0\sin\theta\, t - \frac{1}{2}gt^2$.

i) $\tan\theta = \frac{h}{d} \Rightarrow h = d\tan\theta = 60\sqrt{3}\tan 60 \Rightarrow \boxed{h = 180\,m}$

$y = \tan\theta\, x - \frac{g}{2u_0^2\cos^2\theta}x^2$ άρα όταν $x = d \Rightarrow y = h'$

$h' = \tan\theta\cdot d - \frac{g}{2u_0^2\cos^2\theta}d^2 \Rightarrow h' = 180 - \frac{10\cdot(60\sqrt{3})^2}{2\cdot 60^2\cdot\cos^2 60} = 180 - 60 \Rightarrow \boxed{h' = 120}$

ii) $h - h' = 60\,m$ Για το σώμα με μάζα $m/2$: $h - h' = \frac{1}{2}gt_\Sigma^2$

$\Rightarrow t_\Sigma = \sqrt{\frac{2(h-h')}{g}} = \sqrt{12} \Rightarrow t_\Sigma = 2\sqrt{3}\,s$

$\Rightarrow u_{2y} = g t_\Sigma = 20\sqrt{3}\,\frac{m}{s}$ $(v_{2x} = 0)$

$u_{1x} = u_0\cos\theta = 30\,m/s$ και $u_{1y} = u_0\sin\theta - g t_\Sigma$.

$u_{1y} = 10\sqrt{3}\,m/s$

} Ταχύτητες πριν την κρούση.

Με διατήρηση της ορμής σε κάθε άξονα μπορούμε να βρούμε τις ταχύτητες μετά την κρούση:

1) $P_{x,\text{πριν}} = P_{x,\text{μετά}}$: $m u_{1x} = \left(m + \frac{m}{2}\right)v_{0x} \Rightarrow v_{0x} = \frac{2}{3}u_{1x}$

$v_{0x} = 20\,m/s$

2) $P_{y,\text{πριν}} = P_{y,\text{μετά}}$: $m u_{1y} + \frac{m}{2}u_{2y} = \left(m + \frac{m}{2}\right)v_{0y} \Rightarrow v_{0y} = \frac{2}{3}\left(u_{1y} + \frac{u_{2y}}{2}\right)$

$\Rightarrow v_{0y} = \frac{2}{3}\left(10\sqrt{3} + \frac{20\sqrt{3}}{2}\right) = [illegible] \Rightarrow v_{0y} = 0$

[illegible] είναι στο μέγιστο

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Οριζόντια βολή με $V_{ox} = 20\,m/s \Rightarrow V_{oy} = 0$

$V_x = V_{ox} \quad V_y = gt$

$x = V_x t \quad y = \frac{1}{2} g t^2 \Rightarrow$ όταν $y = h' \Rightarrow t_{ολ} = \sqrt{\frac{2h'}{g}}$

$\Rightarrow R = V_x \sqrt{\frac{2h'}{g}} = 20\sqrt{\frac{2 \cdot 120}{10}}$

$R = 20\sqrt{24} \Rightarrow R = 40\sqrt{6}\,m \Rightarrow A\Gamma = d + R$

$A\Gamma = 60\sqrt{3} + 40\sqrt{6} = 202\,m.$

iii) $\vec{J} = \Delta\vec{P}$

$\vec{J} = \Delta P\,\hat{i} + \Delta P\,\hat{j}$

$= \frac{m}{2}(V_{ox} - 0)\,\hat{i} + \frac{m}{2}(V_{oy} - u_{2y})\,\hat{j} = \frac{m}{2} V_{ox}\,\hat{i} - \frac{m}{2} u_{2y}\,\hat{j}$

$\vec{J} = (20\,\hat{i} - 20\sqrt{3}\,\hat{j})\ kg\frac{m}{s} \Rightarrow |\vec{J}| = \sqrt{20^2 + -(20\sqrt{3})^2}$

$|\vec{J}| = 40\,kg\frac{m}{s} = 40\,N\cdot s$

Άσκηση 2

i) $u = u_0 = 0{,}12\,m/s$

ii) $\left.\begin{aligned} m_\mu u_0 &= m_\mu u_\mu \cos\varphi + m_K u_K \cos\theta \\ 0 &= -m_\mu u_\mu \sin\varphi + m_K u_K \sin\theta \end{aligned}\right\}$ $\tan\theta = \frac{u_\mu \sin\varphi}{u_0 - u_\mu \cos\varphi} = \frac{0{,}04\,\frac{1}{2}}{0{,}12 - 0{,}04\,\frac{\sqrt{3}}{2}}$

$\tan\theta = 0{,}234 \Rightarrow \theta = 13{,}2°$

$\Rightarrow \boxed{m_\mu u_0 \sin\varphi = m_K u_K \sin(\theta+\varphi)}\ *$

$\boxed{\frac{1}{2} m_\mu u_0^2 = \frac{1}{2} m_\mu u_\mu^2 + \frac{1}{2} m_K u_K^2}$ ← διαιρώ τις δυο εξισώσεις

$\frac{u_0^2 - u_\mu^2}{u_0 \sin\varphi} = \frac{u_K}{\sin(\theta+\varphi)}$

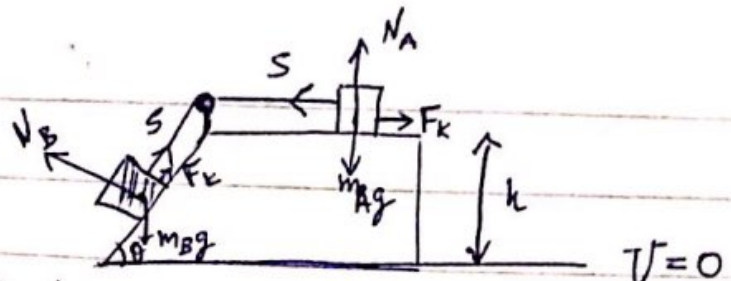
$\Rightarrow u_K = \frac{u_0^2 - u_\mu^2}{u_0 \sin\varphi} \sin(\theta+\varphi) = \frac{0{,}12^2 - 0{,}04^2}{0{,}12 \cdot \frac{1}{2}} \sin(43{,}2°) = 0{,}15\,m/s$

iii) $m_K = \frac{u_0 \sin\varphi}{u_K \sin(\theta+\varphi)} \cdot m_\mu = \frac{0{,}12 \cdot \frac{1}{2}}{0{,}15 \sin 43{,}2°} \cdot 0{,}4 \Rightarrow m_K = 0{,}11\,kg.$

iv) Αποδείξαμε ότι $\theta + \varphi = 90°$

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ΑΣΚΗΣΗ 3



i) Έστω ότι θα κινηθεί δεξιά

Σώμα B : $\Sigma F_y = 0 \Rightarrow N_B = m_B g\cos\theta$

$\Sigma F_x = m_B a \Rightarrow m_B g\sin\theta - \mu_K m_B g\cos\theta - S = m_B a.$

Σώμα A $\Sigma F_x = m_A a \Rightarrow S - \mu_K m_A g = m_A a$

ii) $\Rightarrow a = \left(\frac{m_B(\sin\theta - \mu_K\cos\theta) - \mu_K m_A}{m_A + m_B}\right) g$ $\quad a>0 \Rightarrow$ ΘΕΤΙΚΗ ΚΙΝΗΣΗ.

iii) $S = m_A a + \mu_K m_A g$

iv) $v = \sqrt{2ad} = \sqrt{2gd\left(\frac{m_B(\sin\theta - \mu_K\cos\theta) - \mu_K m_A}{m_A + m_B}\right)}$

2ος τρόπος

$W_{other} + m_B g d\sin\theta + m_A g h = \frac{1}{2} m_A u^2 + \frac{1}{2} m_B u^2 + m_A g h$

$W_{other} = -\mu_K m_B g\cos\theta . d - \mu_K m_A g d$

$\Rightarrow u = \sqrt{2gd\,\frac{m_B(\sin\theta - \mu_K\cos\theta) - \mu_K m_A}{m_A + m_B}}$

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